Project Name:

Team Number: 206
Project Name: Pet Feeder - Light Sensor Board
Team Member Name: Justin Gonzalez
Version: 1.0

A. All Major Components	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Curiosity Nano board	PIC18F57Q43 Curiosity Nano (DM164150)	1.8 - 5.5V	1	23.82	23.82	mA
	Quad op amp (3 buffers + 1 tied off)	MCP6004	1.8 - 6V	1	0.68	0.68	mA
	Sensor dividers, buttons, LEDs, J3 lines (aggregate)	Resistive loads (see Calcs tab)	5V	1	8.83	8.83	mA

B. +5V Power Rail	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Curiosity Nano board	PIC18F57Q43 Curiosity Nano (DM164150)	1.8 - 5.5V	1	23.82	23.82	mA
	Quad op amp (3 buffers + 1 tied off)	MCP6004	1.8 - 6V	1	0.68	0.68	mA
	Sensor dividers, buttons, LEDs, J3 lines	Resistive loads (see Calcs tab)	5V	1	8.83	8.83	mA
					Subtotal	33.33	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	**41.67**	mA
c1. Regulator or Source	+5V linear regulator	L7805ABV-DG	7.5 - 35V	1	1500.00	1500.00	mA
					Total Remaining Current Available on +5V Rail	**1458.33**	mA

D. External Power Source 1	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selecti	Plug-in wall supply (5.5 mm barrel)	BestCH 9V 3A (EA10302 repl.)	120VAC	+9V	3000.00	3000.00	mA
Power Rails Connected to External Power Source 1							
	+5V regulator (input current from 9V)	L7805ABV-DG	7.5 - 35V		47.67	47.67	mA

Total Remaining Current Available on External Power Source 1 **2952.33** mA

E. Battery Life	Component Name	Part Number	Supply Voltage Range		Capacity (mAh)	Required By Regulators	
	Not applicable	Wall supply used			N/A	N/A	
					Battery Life	N/A	

Notes

23.8 mA estimated operating current based on PIC18F57Q43 datasheet Tables 47-2 and 47-3.

MCP6004: Includes all four amplifiers, including the unused amplifier.

Resistive Loads: Sensor dividers, buttons, LEDs, and J3 signals are grouped together. Individual calculations are provided in the Calcs tab.

L7805ABV: Rated for up to 1.5A output. Regulator specifications and thermal estimates are based on the ST datasheet.

J3: Carries logic signals and ground only. The motor and driver are powered separately and included in the teammate's budget.

J2: Alternate 5V bench-supply input that bypasses the regulator. J1 and J2 must not be powered simultaneously.

Power Supply: BestCH 9V/3A adapter supplies the regulator through J1. Verify connector polarity and compatibility before use.